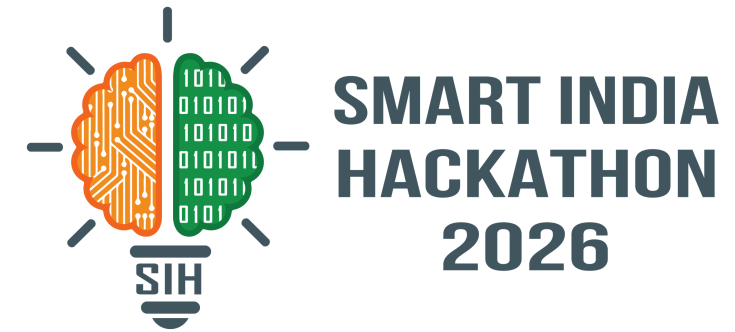


SMART INDIA HACKATHON 2026



Problem Statement ID: SIH26037

Problem Statement Title: Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

Theme: Smart Vehicles

PS Category: Software

Team ID:

Team Name: HardCoders



AI-Powered Road Hazard Analytics, Predictive Maintenance & Driver Safety Navigation

THE GAP WE SAW

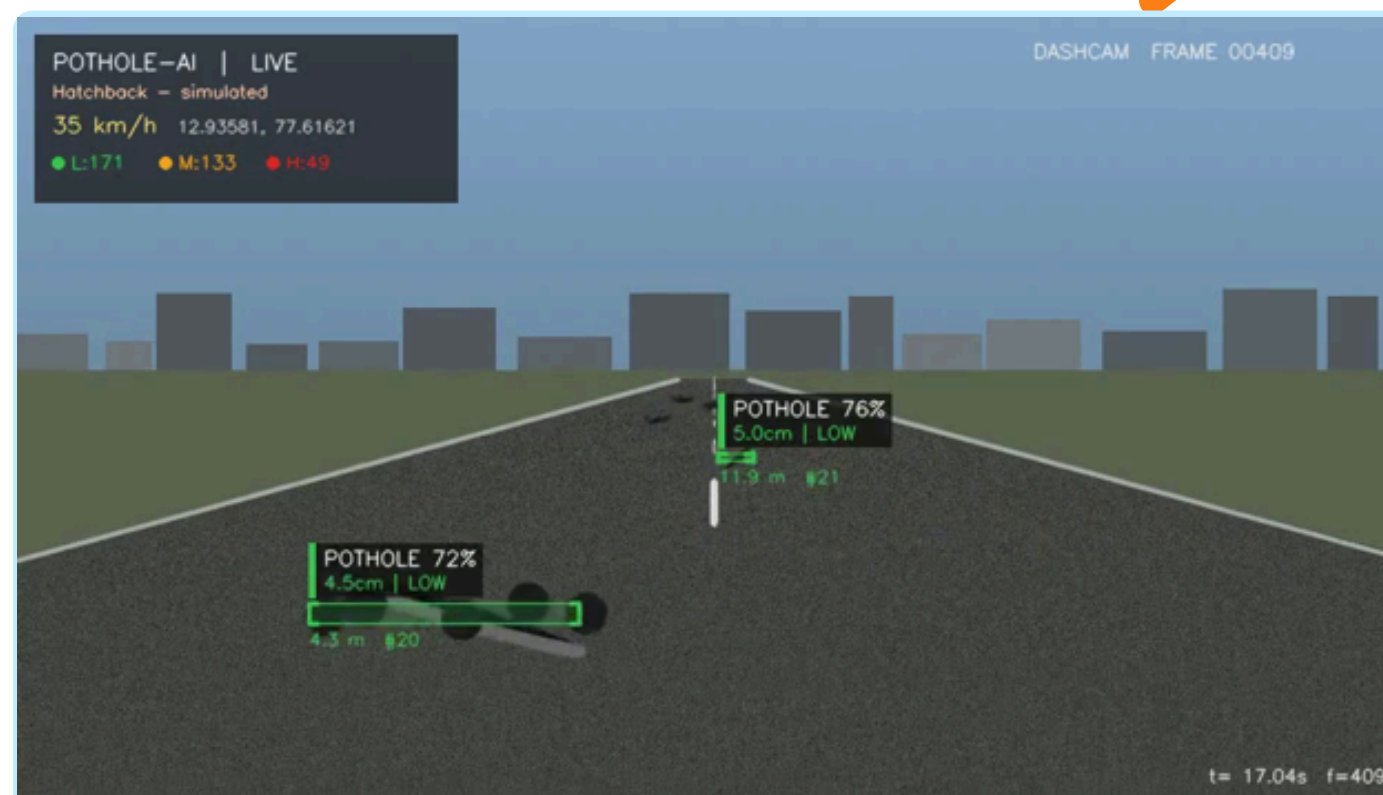
- 1x / yr Typical municipal survey frequency — against a surface that degrades every single day
- 45-90 D Complaint → inspection → tender → repair. Replace with your own city's RTI-sourced figure
- 0 Cheap, objective checks that a repair was actually executed before the invoice clears

How Dodge Closes The Gap

- Removes duplicate frame-by-frame detections.
- Converts pixels into road measurements.
- Scores impact for the specific vehicle sends actionable GPS markers to drivers & repair teams.

BEYOND DETECTION

- Vehicle aware Impact Severity Index.
- Pothole projected to it's true road position.
- DBSCAN merges repeated observations.
- One drive produces both safety alerts and repair intelligence.



Working Prototype - Live Dashcam HUD

Proposed Solution

DODGE converts an ordinary dashcam drive into a real time road hazard intelligence system:

Detect>measure>score>
de-duplicate>map>alert

FROM ROAD TO DATA

How DODGE turns a moving vehicle into a road-observation system



THE TECH BEHIND DODGE

Python

scikit-learn/
DBSCAN

OpenCV

Folium

YOLOv8

NumPy/
Pandas

GNSS/
GPS

THE DODGE PIPELINE

DETECT

YOLOv8/fallback
pixel bounding box

MEASURE

Ground plane
projection
Area+Range

SCORE

Impact Severity
Index
vehicle+speed

GEOTAG

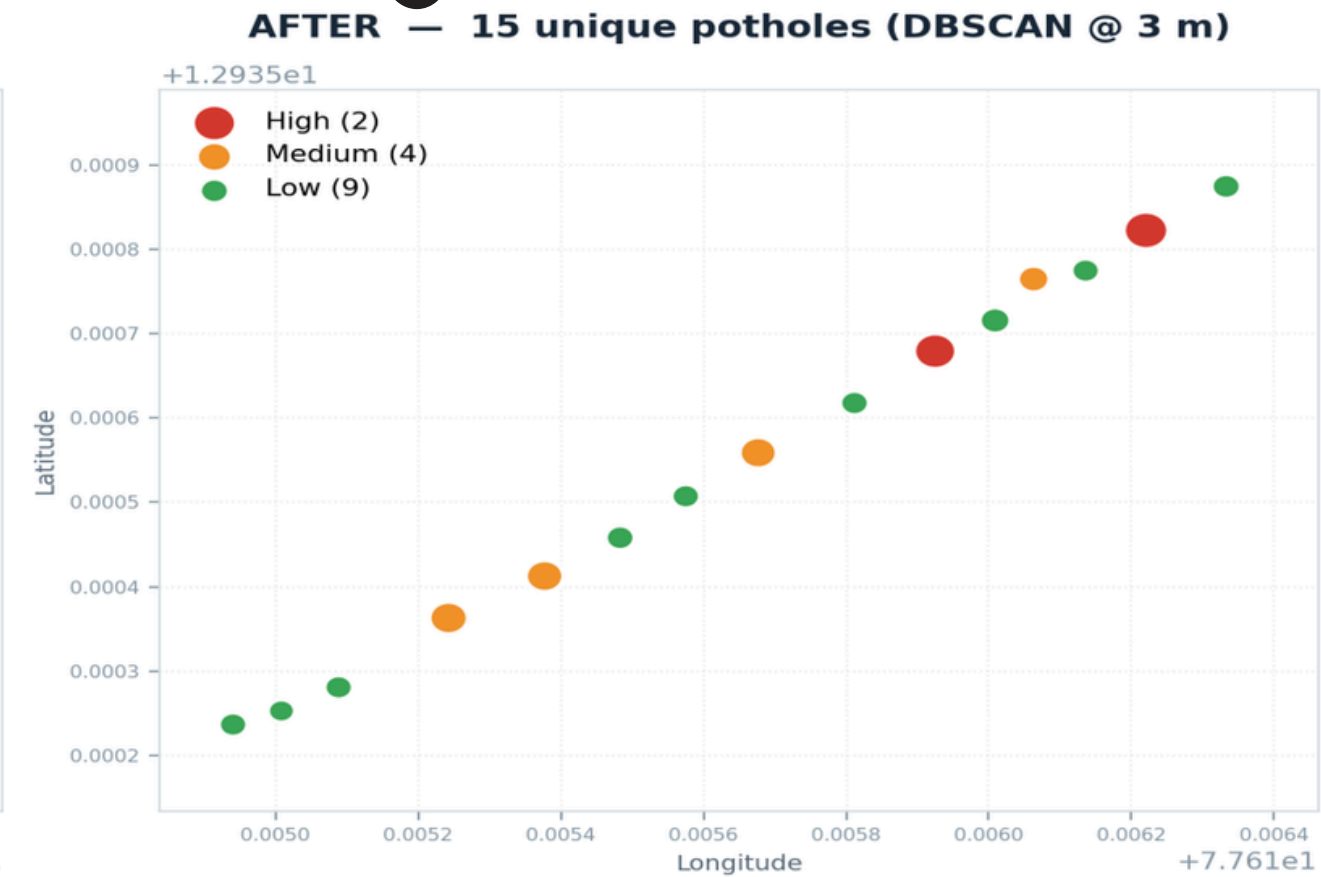
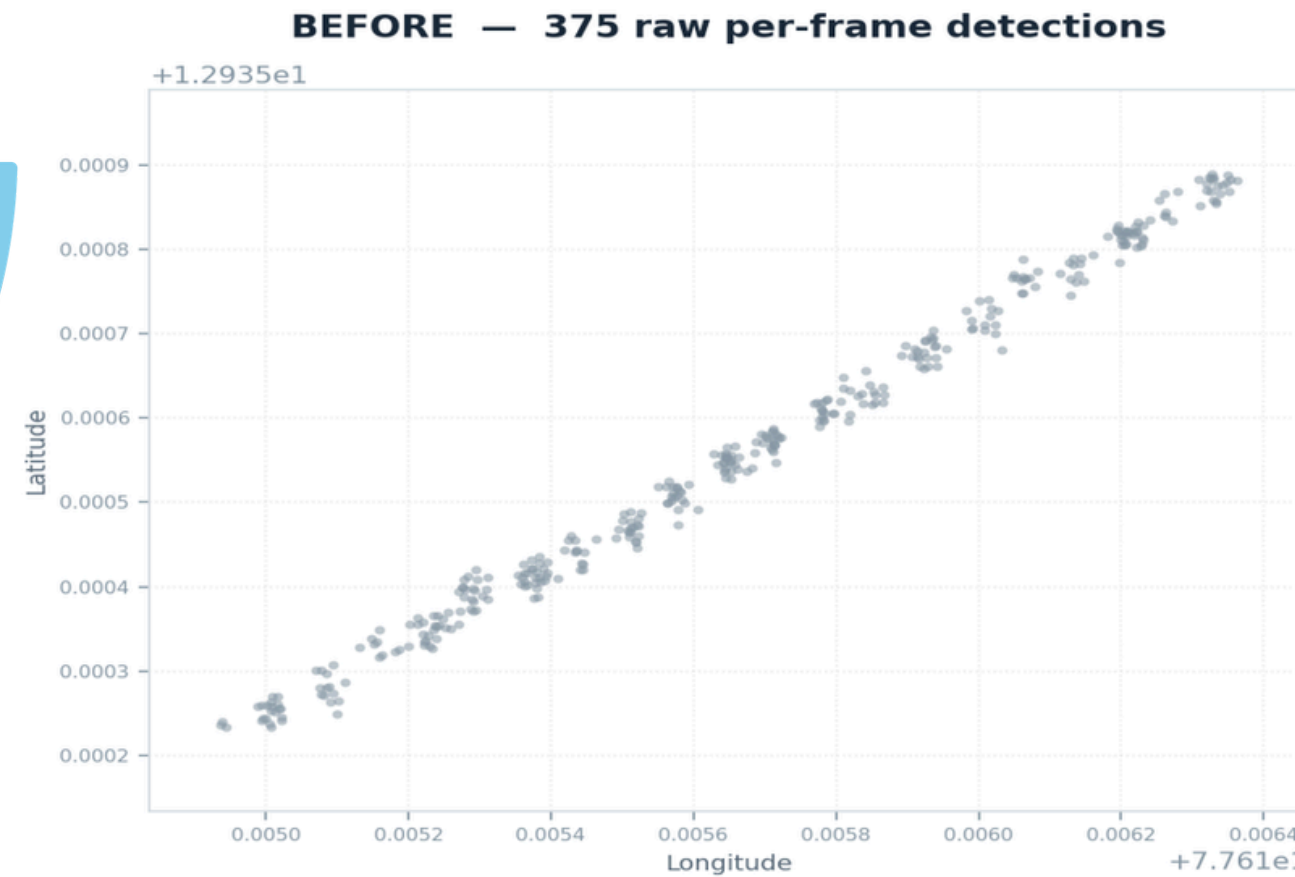
GNSS+Range+
Heading
Road position

CLUSTER

DBSCAN@3m
unique hazards

MAP/ALERT

Folium repair map
HUD warning+log



FROM PROTOTYPE TO ROAD

Technical Feasibility

- Uses mature, accessible tools: YOLOv8, OpenCV, scikit-learn, FastAPI/Folium
- CPU only prototype already runs in real time
- Detector can be swapped without changing analysis stages

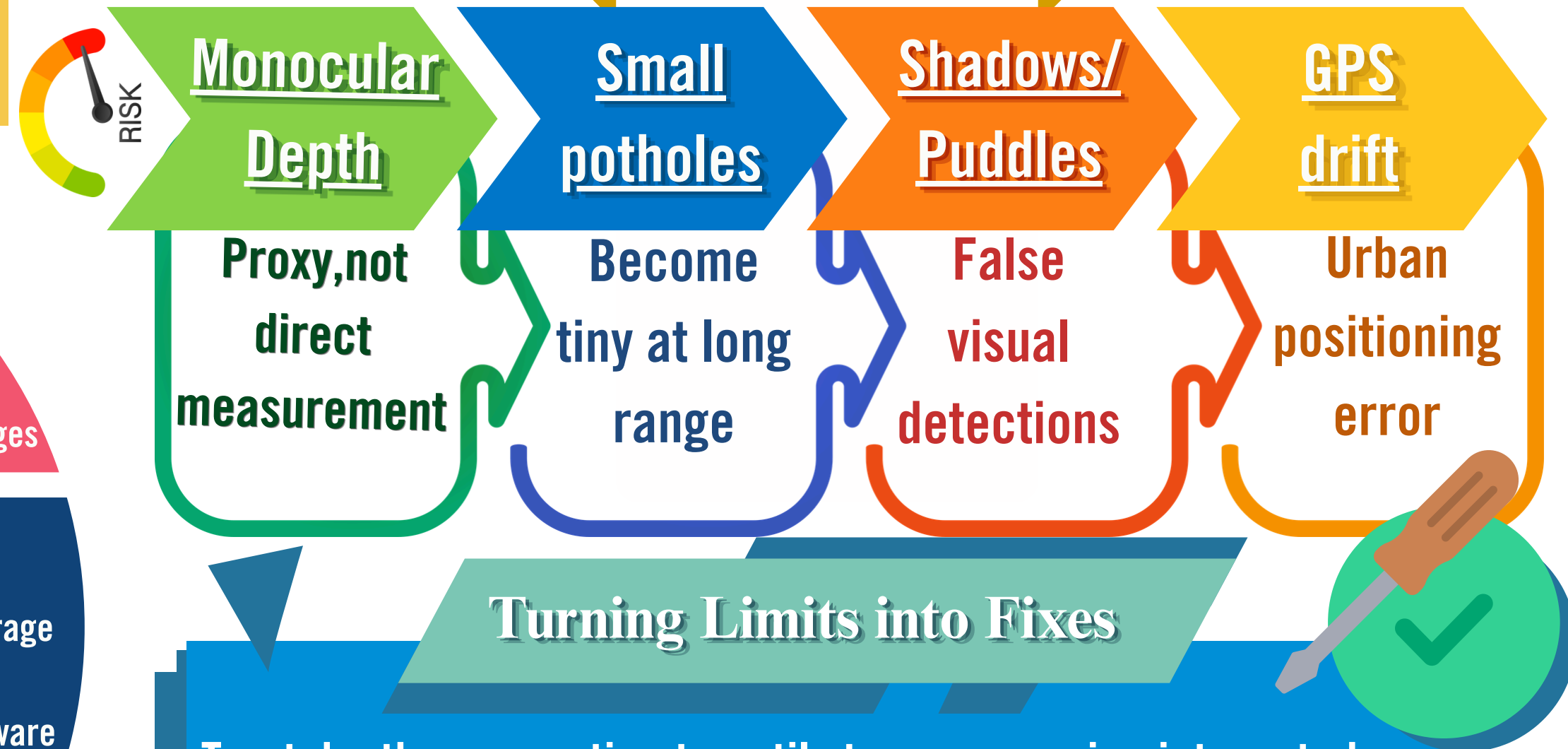
Deployment Feasibility

- Low cost dashcam/ Raspberry Pi tier for high count coverage
- Higher performance edge tier can add stereo/depth later
- Same event format supports mixed hardware

Economic Viability

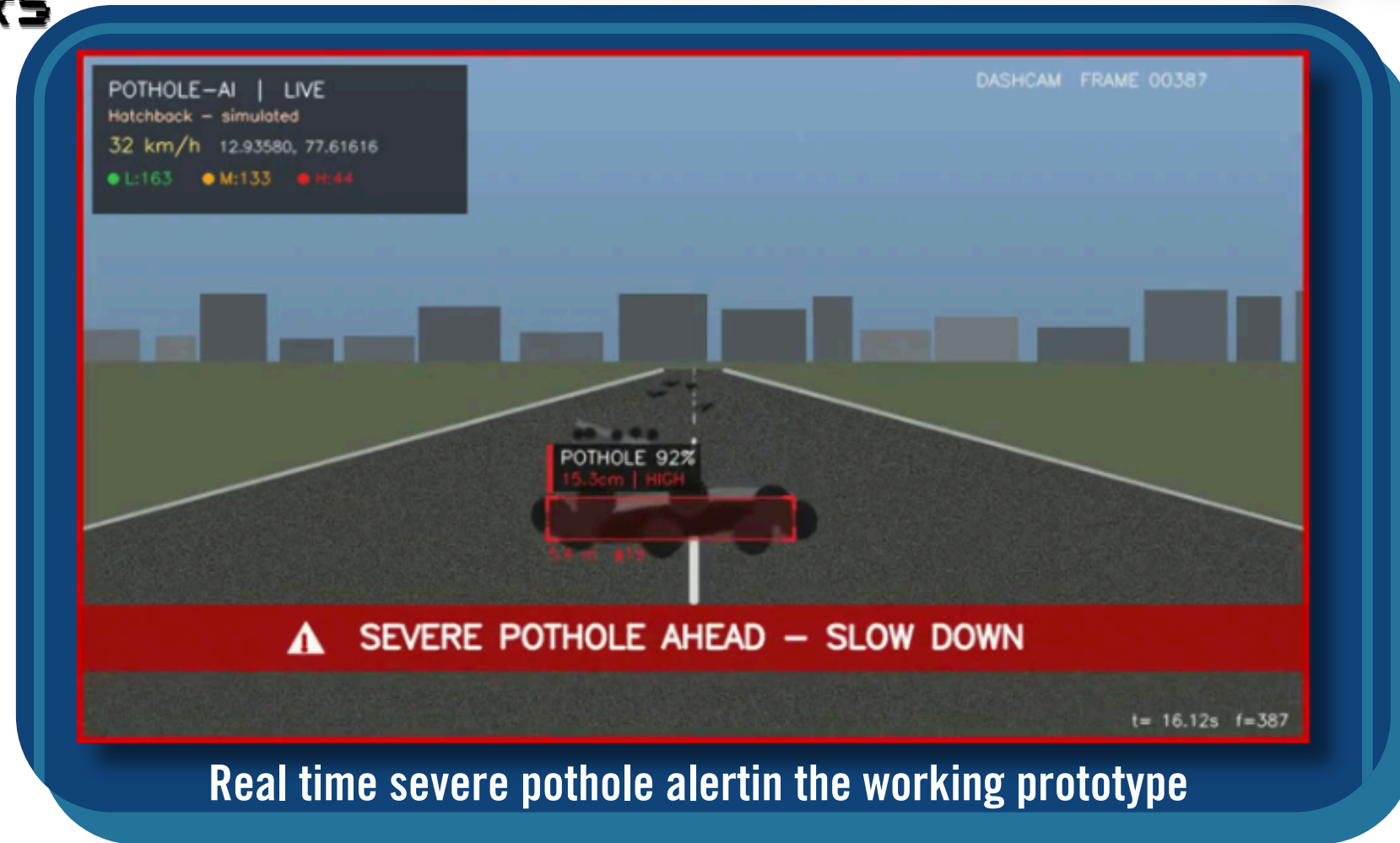
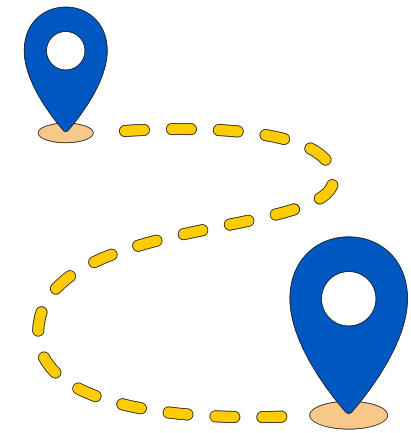
- One drive contribution reusable road data
- De-duplication avoids repeated repair records
- Target users: Fleets, logistics operators & ULBs

What we are up Against



Turning Limits into Fixes

- Treat depth as an estimate until stronger sensing integrated.
- Project potholes-not the vehicle-before clustering.
- Use repeated observations to strengthen confidence.
- Train with road specific hard negatives: shadows, patches, puddles.
- Keep modules independent so better sensors/modules can drop in.



Real time severe pothole alert in the working prototype

Benefits from DODGE

- SOCIAL: FASTER HAZARD AWARENESS AND SAFER MOBILITY
- ECONOMIC: LESS VEHICLE DAMAGE AND MORE TARGETED MAINTENANCE
- OPERATIONAL: 375 NOISY OBSERVATIONS CAN BECOME 15 USABLE REPAIR MARKERS
- SCALABLE: EVERY PARTICIPATING VEHICLE BECOMES A MOVING DATA COLLECTOR
- ACCESSIBLE: PROTOTYPE RUNS ON CPU WITHOUT A GPU

1

For Municipal/Repair Team

- Prioritised GPS markers
- Fewer duplicate reports
- Repair planning based on actual hazard locations



For Driver

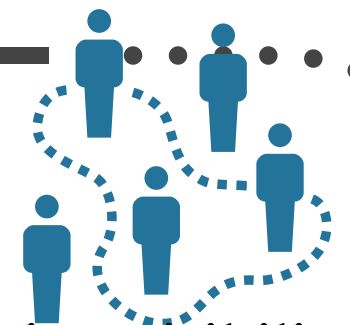
- Early warning before a high-impact hazard
- Vehicle specific severity
- Safer speed/lane decisions



2

For Citizen

- Safer daily travel
- community road condition visibility
- Data generated during normal journeys



3



Datasets & Research

RDD2022 - A Multi-national Image Dataset for Automatic Road Damage Detection (Deeksha Arya et al.) - IEEE BigData / Geoscience Data Journal

- **Research Paper** - <https://arxiv.org/abs/2209.08538>
- **Dataset**- [https://figshare.com/articles/dataset/RDD2022-The multi-national Road Damage_Dataset/21431547](https://figshare.com/articles/dataset/RDD2022-The_multi-national_Road_Damage_Dataset/21431547)

- Ultralytics YOLOv8 - Object Detection & Real-time Inference -

Models & Tools-

- <https://docs.ultralytics.com/models/yolov8>

- MiDaS - Monocular Depth Estimation-<https://github.com/isl-org/MiDaS>

- Geospatial Mapping (Leaflet.js) - <https://leafletjs.com/>

Government & Standards-

- Ministry of Road Transport & Highways (MoRTH) - <https://morth.nic.in>
- IRC SP:88 - Guidelines for Road Maintenance Management System
- OpenStreetMap (OSM) - <https://www.openstreetmap.org>

Libraries & APIs-

- OpenCV <https://opencv.org>
- Google Maps API - <https://developers.google.com/maps>
- FastAPI - <https://fastapi.tiangolo.com>

